Enhanced photoluminescence from GaN grown by Lateral Confined Epitaxy

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ABSTRACT

Lateral Confined Epitaxy (LCE) of GaN on Si substrates results in a reduction of thermal crack density with decreasing the lateral dimensions of the growth pattern. Below a critical size, crack-free GaN on Si is obtained. The intensity of band-to-band photoluminescence (PL) peak in LCE GaN is strongly enhanced with respect to uniformly grown GaN on Si. The present study rules out the effect of crack density, internal reflections (microcavity effects), as well as enhanced light extraction efficiency, and excitation or emission through preferred facets (shape effects) as the main factor in PL enhancement. It is shown that the reduction in threading dislocation density (TDD) along the edges of the LCE patterns improves the luminescence efficiency. The relative increase in high quality material (low TDD) with the reduction of LCE unit size is, thus, the main reason for the enhanced PL intensity.

Keywords: GaN on Si, Lateral Confined Epitaxy, Photoluminescence.
PACS: 81.05.Ea, 81.10, 81.10.Bk, 81.15.Gh, 78.55, 81.40.Tv.

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INTRODUCTION

When GaN is epitaxially grown on Si substrates it suffers from dense cracking due to the large difference in thermal expansion coefficient between GaN and Si. Recently, we have shown that Lateral Confined Epitaxy (LCE) can be used for controlling the crack density on GaN mesas of finite lateral dimensions.[1, 2] For growth units (mesas) below a certain critical dimension, crack-free GaN on Si(111) can be obtained.[1]

For the implementation of LCE, the Si substrates are first patterned into square mesas, separated by shallow trenches. The subsequent growth is not selective (material is deposited both on the mesas and in the trenches). However, the trenches are effective in mechanically de-coupling neighboring mesas. Thus, strain release at the periphery of the mesa takes place, resulting in a reduction in density of thermal cracks with decreasing the size of mesas used for LCE.

Apart from the morphological improvement in the material, it was also found that the band-to-band (BB) photoluminescence (PL) efficiency increases dramatically with

reduction of mesa size in LCE GaN/Si.[1] Here, we report on experiments performed in order to study the effect of various factors on PL efficiency in GaN.

Room-temperature photoluminescence (PL) intensity of the BB peak of a direct-gap semiconductor can be enhanced by a reduction in the density of the non-radiative centers, i.e. an improvement of the material quality. It may also be modified by changes in the shape of the excited material, via microcavity effects, or changes in light extraction efficiency, due to reduction of total internal reflection from the semiconductor/air interface.[3]

Several factors should be considered as possible causes of the enhancement of the PL intensity in LCE layers. These factors may be divided into the two categories, namely (1) material quality and (2) shape effects.

The first category includes (*i*) thermal cracks, and (*ii*) extended defects, including threading dislocations (TD's) or grain boundaries. These are considered "internal surfaces" where PL intensity is reduced due to non-radiative recombination through surface states.[4]

The second category (shape effects), may include (*i*) the amount of excited material, which may change in different mesa/trench arrays, since in fact, less mesa material exists in the arrays with smaller patterns; (*ii*) reflection from the pattern boundaries. The reflection may then be of the exciting radiation or of the emitted radiation, which may be enhanced due to microcavity effects;[5] (*iii*) the light extraction efficiency in different areas (a rougher surface widens the extraction cone).

The main challenge in this study is the distinction between the different effects of material and shape properties on PL and the determination of the predominant factor in enhancing LCE-GaN emission.

EXPERIMENTAL

GaN layers were grown by low pressure (100 Torr) MOCVD in a vertical Thomas Swan reactor. Precursors were Trimethylaluminium, Trimethylgallium and Ammonia. The GaN/Si samples contained a ~50 nm AlN buffer layer while the GaN/sapphire had a GaN buffer layer, separated from the main GaN layer by a thin insulating AlGaN layer.

Several effects were examined in this work, using different experiments. First, the influence of thermal cracks on PL intensity was investigated. In order to do that, LCE-GaN was grown on Si and on sapphire, using the same set of patterns. In contrast to GaN on Si, the LCE GaN on sapphire is crack-free for all pattern sizes. If thermal cracks strongly affect the PL intensity, an enhancement in the PL with reducing pattern size is expected in GaN-on-Si, but not in GaN on sapphire. In addition to that, a sample of GaN grown uniformly on a Si substrate was patterned after growth (pg-GaN/Si), with the same pattern sizes, to test the effect of the mesa size with a constant material quality (i.e. crack density and TD density). In this sample the influence on PL intensity should originate only from shape effects.

Si and sapphire substrates for LCE were patterned by dry etching prior to growth. The patterns were square mesas of varying sizes a, in the range 5 μm ≤ a ≤ 100 μm, separated by trenches 0.5-0.6 μm deep and 2 to 4 μm wide.

In order to study the influence of morphology on PL, a GaN film with a spatially varying pit density was produced. To do that, the strong influence of growth temperature (T_g) on the film quality was used. First, the growth of GaN nucleation

layer and its annealing were performed. Then, prior to growth of GaN main layer, the substrate was mounted on a wedge, to induce a spatial gradient in thermal coupling to the heating element. This resulted in a spatial change of T_g during growth of ~ 50°C, and a consequent spatial gradient in density of pits (more pits in areas of lower T_g). This sample was used to correlate PL intensity with pit density.

PL studies were performed at room temperature, using a HeCd laser (~10^3 mW/cm^2) as excitation source. The emitted radiation was dispersed by a 0.3 m monochromator and detected by a photomultiplier. Structural characterization was performed by plan-view and cross-sectional transmission electron microscopy (TEM). The two beam and weak beam diffraction contrast as well as high resolution TEM were carried out using a JEOL 2010 instrument, operating at 200 keV. Morphology studies were performed by high resolution scanning electron microscopy (SEM) at 10 keV. Crystalline orientation was studied by X-ray diffraction (XRD). The peak width of $(0002)_{GaN}$ rocking curve was measured by high resolution XRD, and all GaN reflections were analyzed by powder diffraction θ-2θ scans.

RESULTS AND DISCUSSION

At the wavelength of 325 nm, which was used to excite the layers, GaN absorption coefficient was evaluated as $\alpha \sim 2.7\times10^5$ cm^{-1}.[6] This value corresponds to an absorption depth of ~370 Å. All GaN layers examined, including LCE layers, both on mesas and in trenches, are at least 1 order of magnitude thicker, over the whole excited area. Thus, the amount of excited material is similar for all the experiments.

Figs. 1(a) and 1(b) show the normalized BB peak intensity of the LCE GaN/Si and GaN/sapphire layers, respectively. The peak intensity clearly increases with the reduction of LCE pattern size on both substrates, however the improvement factor is smaller in the GaN/sapphire samples. A study of the $(0002)_{GaN}$ peak in XRD showed no change in the full width at half maximum (FWHM) with pattern size.

Since GaN/sapphire does not exhibit thermal cracking, the enhancement in PL intensity with reduction of pattern size on these samples rules out thermal cracks as a major mechanism that influences PL efficiency. However, the enhancement in PL intensity for small pattern sizes ($a \leq 30$ μm) is significantly larger in GaN/Si than that of GaN/sapphire. This is the range in growth pattern size in which thermal crack density in GaN/Si is drastically reduced.[2] Thus, the fact that the improvement in GaN/Si is larger than that of GaN/sapphire may indicate that the reduction of thermal crack density could account for ~10 % of the PL enhancement.

Fig. 2 shows plan-view SEM micrographs of the smallest patterns (~5 μm) in the LCE-GaN/Si [2(a)], LCE-GaN/sapphire [2(b)] and post-growth (pg)-GaN/Si [2(c)]. The micrographs show GaN on mesas and in adjacent trenches. The LCE-GaN grown on top of the mesas is of high quality. The GaN in the trenches seems to be of similar quality. In fact, on the sapphire substrate bridging occurs between GaN on mesas and in the trenches. However, more pits are observed in the trench material due to lower growth rate, arising from the substrate topography. The shape of the GaN on mesa is different between LCE- and pg-systems. While in the LCE-GaN a clear faceting is observed, the pg-GaN/Si has a more rounded shape, which is a result of a non-optimized dry etching process.

The material quality in the pg-GaN/Si is constant in all pattern sizes, since all patterns originate from the same film. Thus, only size or geometric effects may influence the PL intensity. PL measurements of the pg-GaN/Si showed neither systematic nor significant difference in intensity between the patterned and unpatterned areas, or between different pattern sizes. This rules out the pure pattern size dependence of PL intensity.

Figs. 3(a) and 3(b) show plan view TEM images of 20 μm and 100 μm patterns, respectively, taken from the center of the nesa. The black dots and the short black lines are top surface terminations of TD's, which originate at the substrate/layer interface. The areas of different contrasts show adjacent sub-grains (sg) with a small tilt of the c-axis in respect to one another. They enable the estimation of sg size in these patterns. It is clearly seen that the sg in the 20 μm pattern are about 2 times larger than in the 100 μm patterns, i.e. less grain boundaries are present in the smaller patterns. A more detailed description of the microstructure is found in Ref. 7.

The effect of the sg size on PL is complicated to estimate. Since sg boundaries are composed of dislocations which thread from the substrate to the upper surface, (small angle sub-grain boundary or dislocation walls), and since these TD's are mainly observed at the sg boundaries (Fig. 3), the sg boundaries may be included in the calculations concerning the TDD at the surface. Ponce *et al.*[8] studied a non-ideal layer which contained hexagonal hillock-like crystallites, and showed that the main part of the BB emission originates from the interior of GaN crystallites, i.e. from the low TDD material. The BB intensity at the grain boundaries (between the hillocks), i.e. from the high TDD material, is lower.

Fig. 4 shows cross-sectional TEM images, taken at g = 0002, revealing the TD's in LCE GaN/Si. Fig. 4(a) was taken near the center of a 20 μm pattern and Fig. 4(b) was taken close to the edge of the pattern. The TDD in the top GaN surface is evaluated as $\sim 10^{10}$ cm^{-2} across the pattern and it decreases to $\sim 10^{9}$ cm^{-2} towards the edge. The area of which the reduction of TDD is observed is ~1-2 μm of the pattern periphery in all pattern sizes. Fig. 4(c) shows the threading dislocations close to a hexagonal pit in the GaN pattern (often termed "V-defect"). The hexagonal pits expose nearly singular $\{1\,\bar{1}\,01\}$ planes. It is seen that the TDD is lowered in this region (~200 nm from each pit wall) and that the inner pit surface is free of TD's. This phenomenon is explained in detail in Ref. 7. Fig. 4(d) shows the trench material, having deep pits between crystallites of pyramidal shapes. The top surface of the crystallites is of very low TDD.

The TDD reduction in the vicinity of pit walls suggests that when the pit-density is higher, the TDD between pits should be lower, as seen in Fig. 4(d), and the relative volume of material with higher crystalline quality should be larger. Thus it is expected that GaN films with larger pit density will exhibit higher BB efficiency. This issue was examined by the sample which was grown under a temperature gradient. Figs. 5(a) and 5(b) are plan-view SEM micrographs of the high and low temperature areas, respectively, of this sample. The high temperature region is almost completely flat with low density of hexagonal pits, while the low temperature region shows high pit-density. The calculated pit density for different regions along the temperature gradient line, and the corresponding BB peak intensity are shown in Fig. 5(c). The PL intensity gets >200 times higher towards the high pit-density region.

The sharp enhancement in PL intensity in the GaN with high pit density [Fig. 5(b)] is in good agreement with the results of Bidnyk *et al.*[9], who found that imperfections in the epi-layer, as well as the vicinity to sample edges, enhance the PL intensity and even produce stimulated emission. Bidnyk *et al.* explained it in terms of defect walls which serve as scattering centers for the photon flux which is parallel to the layer.

Another explanation for the emission enhancement may be the improvement of light extraction efficiency from the film textured top surface, i.e. from pits or pattern edges. However, an improved extraction could lead to an intensity enhancement of no more than one order of magnitude, at optimal conditions, which include strong reflection from the bottom interface and a nanometer scale dense texture of the top surface,[3] both of which do not exist in our layers. Therefore, it cannot be the dominant effect responsible for the ×200 intensity enhancement.

From the above discussion, it is tempting to conclude that the relative volume of low TDD in the vicinity of pit sidewalls and pattern edges is the main factor controlling PL efficiency, since it increases significantly with pit density and with the reduction of pattern size. However, a larger density of edge and pit sidewalls increases the volume of material excited through certain (e.g. $\{1\bar{1}01\}$) crystallographic planes, which may have influence on the measured PL intensity. To study this possibility it would be desirable to produce a GaN sample which combines the following two properties: (*i*) low TDD, and (*ii*) random faceting. If such a sample is produced, and if its PL emission is stronger than that of plain GaN/Si, it should be possible to conclude that TDD is indeed the main factor influencing PL intensity.

Since heteroepitaxy of GaN on foreign substrates results in compromising film crystalline quality, i.e. with high TDD, the opposite route, namely the synthesis of a film of polycrystalline GaN was adopted here.

A polycrystalline GaN film was grown under conditions far from optimal on a Si(100) substrate. Si(100) has no single preferred orientation relationship with GaN epitaxial films. In the polycrystalline GaN film, seen in Fig. 6(a), there are small crystallites, with the largest of the order of ~1 μm. XRD showed that the crystallites are not aligned with any preferred orientation. The crystallites may be easily removed from the substrate by scraping, since most of them are loosely attached to it. This non-epitaxial growth minimizes stresses originating from thermal or lattice mismatch, and therefore the TDD should be lower than the typical values found in epitaxial films. It also assures that the crystallites are neither exited from, nor emit through a certain, single direction or plane.

Fig. 6(b) is a comparison between the PL intensity of this sample and that of a standard, flat, nearly single-crystalline GaN/Si sample. The BB peak intensity is ~100 times stronger for the polycrystalline film. We conclude from this measurement that the most significant effect on the PL intensity is the TDD, i.e. for a certain amount of excited material it is the ratio between low and high TDD regions under excitation.

It has to be noted, that although PL intensity is much stronger in polycrystalline GaN, the expected electroluminescence (EL) of such a film would be of lower efficiency, unless the current injection difficulties through the unordered crystallite boundaries are solved.[10]

The reason for PL enhancement in the lower TDD regions may be due to an improvement of minority-carrier lifetime. The carrier diffusion length increases with the distance between adjacent dislocations.[11] This, in turn, reduces the probability for a non-radiative recombination, and increases the radiative efficiency.

Returning to the LCE, we concluded from Fig. 4 that the relative amount of high quality material is strongly influenced by LCE pattern size or by pit density. Thus, the PL intensity sharply changes with these parameters.

We propose here a simple model to explain the PL intensity dependence on LCE pattern size. Assume that the square mesas in our LCE samples are $a \times a$ in size, separated by trenches of width L [Fig. 7(a)]. Each mesa has a region along its periphery, of width u, with lower TDD. The PL efficiency in the pattern center, the trench and the pattern periphery are η_1, η_2, and η_3, respectively. In a PL experiment with excited area A, we may distinguish among the center pattern area A_c, the pattern periphery area A_u, and the area covered by trench material A_t.

Since the average TDD is highest in the pattern centers and lowest in the pattern edges, it is plausible to expect that $\eta_1 < \eta_2 \leq \eta_3$.

For simplicity, we assume a square excitation area of length $\sqrt{A}$, which corresponds to N times the period $L + a$. Thus the three different areas are:

$$A_t = A - N^2 a^2 = N^2 L(L+2a) \tag{1}$$

$$A_u = N^2 \cdot 4u(a-u) \tag{2}$$

$$A_c = N^2 (a-2u)^2 \tag{3}$$

The total intensity emitted from the surface is

$$I = \eta_1 A_c + \eta_2 A_t + \eta_3 A_u \tag{4}$$

Replacing N^2 with $A/(L + a)^2$, and the areas with Eqs. (1)-(3), gives the total intensity factor

$$\eta = I / A = \begin{cases} \left(\eta_1 (a-2u)^2 + \eta_2 L(L+2a) + \eta_3 \cdot 4u(a-u)\right) / (L+a)^2 & , a > u \\ Max\left\{\eta_2, \ \left(\eta_2 L(L+2a) + \eta_3 \cdot a^2\right) / (L+a)^2\right\} & , else \end{cases} \tag{5}$$

In our samples a is known, L may be 2-4 μm and u was found by TEM cross-sections to be ~1μm. The intensity factors are unknown. Without loss of generality, we assume $\eta_1 = 1$ for flat surfaces, and that η is inversely proportional to the TDD. Thus, for an edge, where the TDD is lower by one to two orders of magnitude, $\eta_3 \sim$ 10-100 (for the model $\eta_3 = 30$ was chosen as a rough estimate). In the trench, η_2, may take any value between 1 to η_3, depending on the ratio of low to high TDD areas. Fig. 7(b) shows the applied model, compared with the PL results on LCE. The error in PL intensity measurements was estimated as ~20%.

The stronger pattern size effect in the GaN/Si samples, when compared to the GaN/sapphire samples was aforementioned as a possible influence of the thermal cracks induced only in GaN/Si. However, it may also be attributed to the smaller density of pits in the GaN grown in trenches on sapphire (see Fig. 2), and therefore to a smaller η_2. For the model, $\eta_2 = 4$ was estimated for GaN/Si and $\eta_2 = 1$ for GaN/sapphire.

The proposed model is oversimplified, and it is not claimed that it can be used for quantitative predictions of PL enhancement. However, it seems that the rough estimate is in qualitative agreement with the experimental results, and it predicts an optimum in PL intensity for a 5.0 μm pattern for Si and 6.0 μm for sapphire. It also predicts a decrease in intensity toward smaller patterns, where the total pattern area reduces to zero.

A similar model may be applied to explain the PL intensity change with the density of pits, where at a very high pit-density the pits overlap, and the overall high-quality area saturates and may decrease.

CONCLUSIONS

The intensity of band-to-band photoluminescence in LCE GaN/Si is strongly enhanced when compared to that of unpatterned GaN/Si layers. The enhancement increases with decrease in LCE pattern size.

The improvement of BB peak intensity in small LCE GaN patterns arises from a relatively higher amount of high-quality material with low TDD, mainly in the vicinity of the free pattern edges in the mesa material. A smaller part of the improvement may be attributed to two effects: a similar reduction in TDD in the trench material, close to pit walls and grain edges, and the reduction of thermal crack density in small patterns of LCE-GaN/Si.

ACKNOWLEDGMENTS

The authors are grateful to Mrs. S. Yofis for technical assistance. This research was supported by the Israel Ministry of Science, Culture and Sports, through an Infrastructure Project; by the Kidron Fund for Microelectronics at the Technion; and by the Fund for the Promotion of Research at the Technion. YG and JS wish to acknowledge support from the Israel Science Foundation, Grant # 243/99-2.

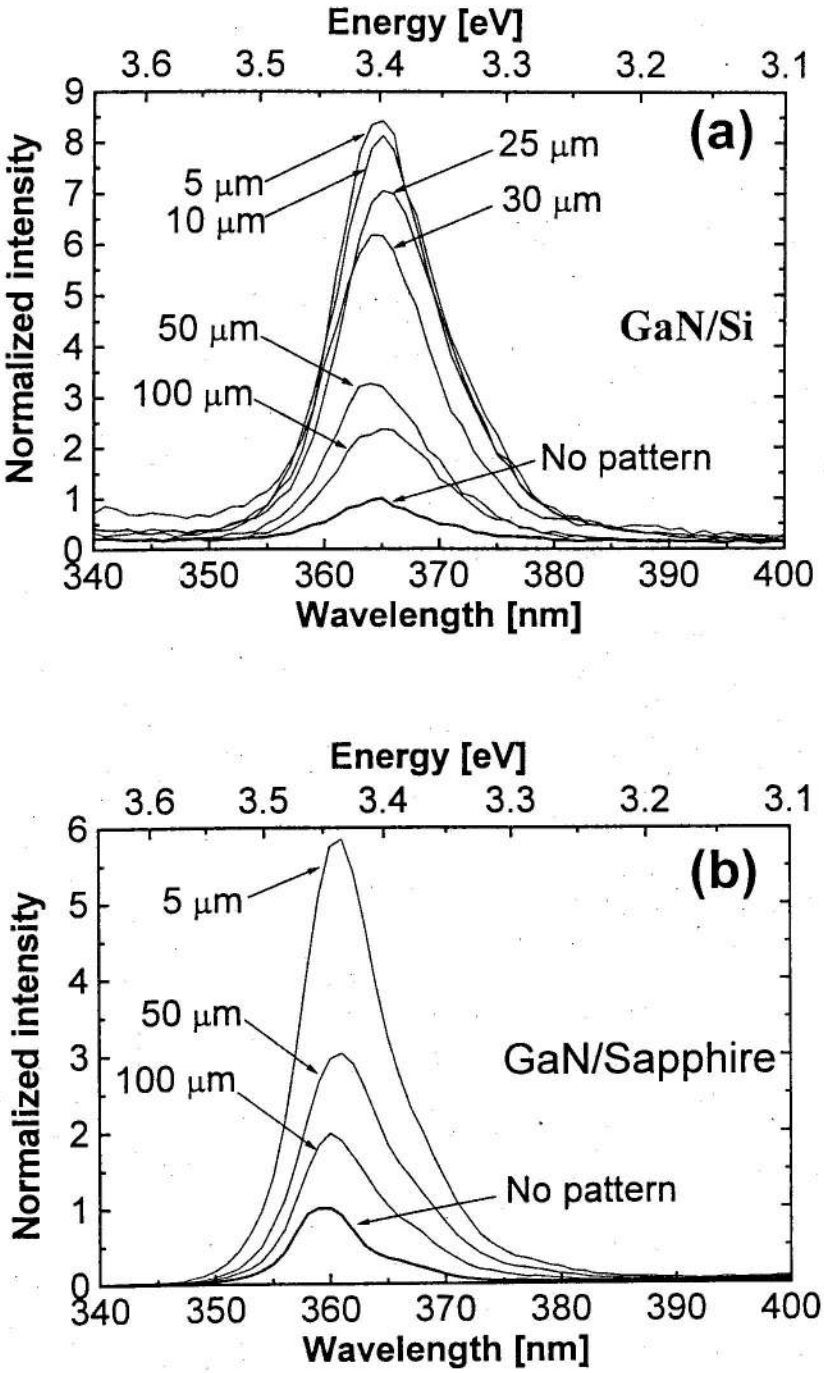


FIG. 1. BB peak intensity of LCE patterns of (a) GaN/Si; (b) GaN/sapphire, for different pattern sizes, normalized to unpatterned layer peak intensity.

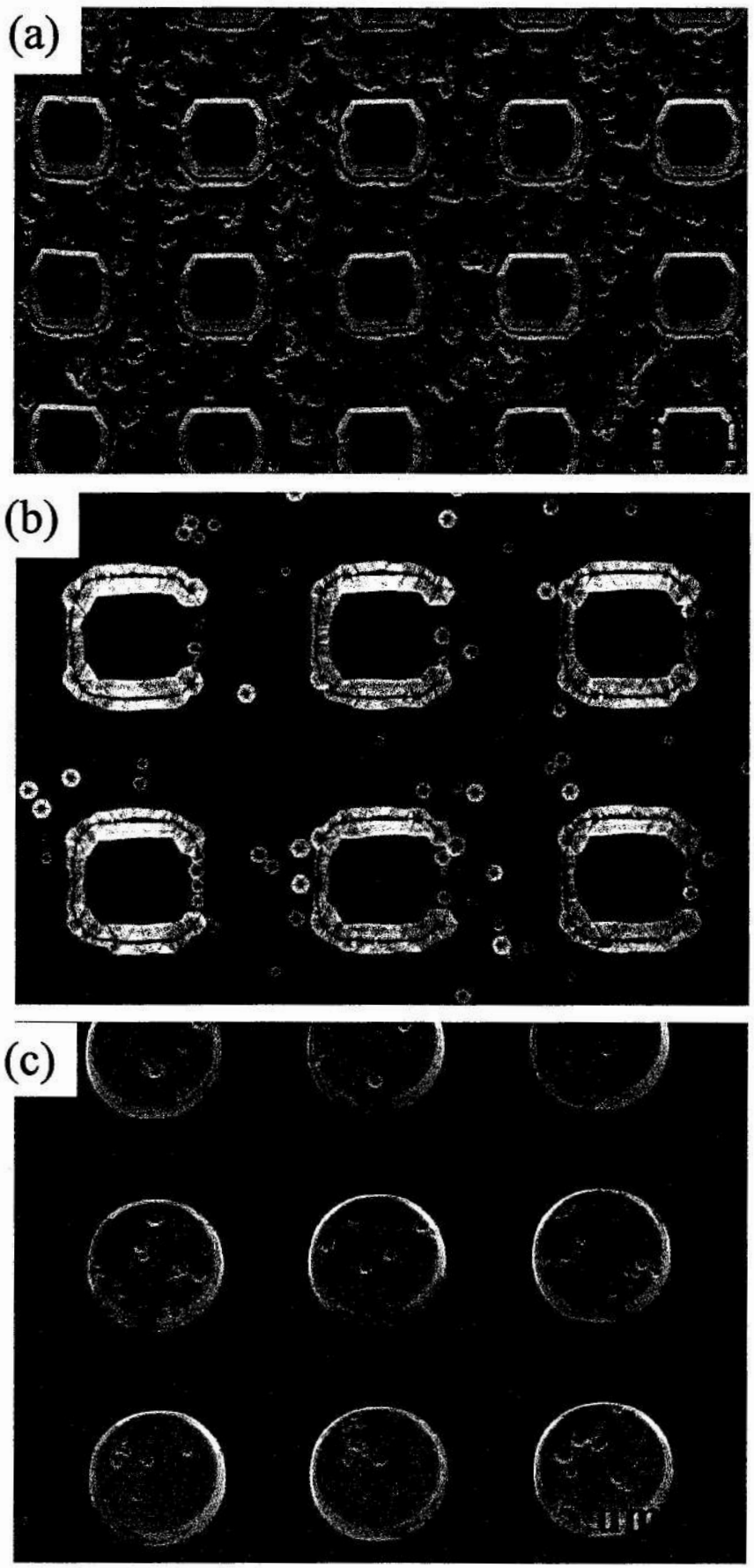


FIG. 2. Plan-view SEM micrographs of the smallest patterns (~5 μm) and adjacent trenches in (a) LCE GaN/Si, (b) LCE GaN/sapphire, and (c) post-growth (pg)-GaN/Si.

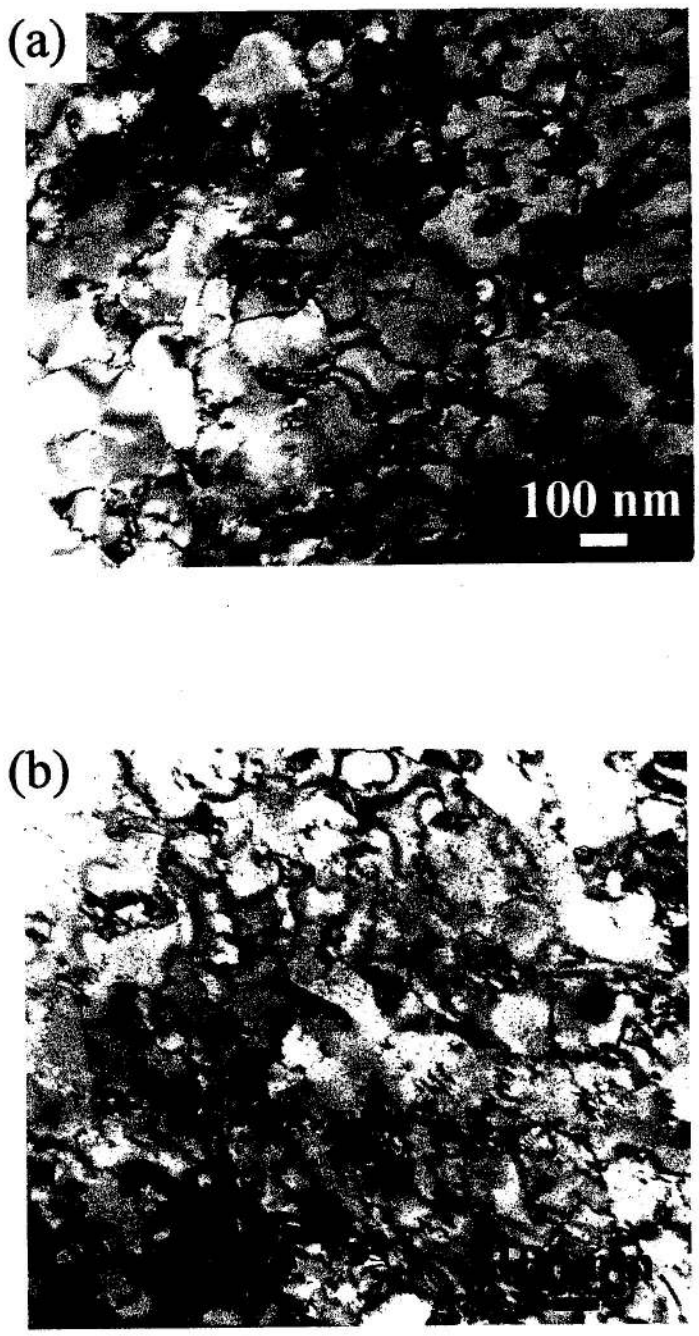


FIG. 3. Plan-view TEM images of (a) 20 μm, (b) 100 μm patterns at their centers. The black dots and the short black lines are TD's intersecting the top surface.

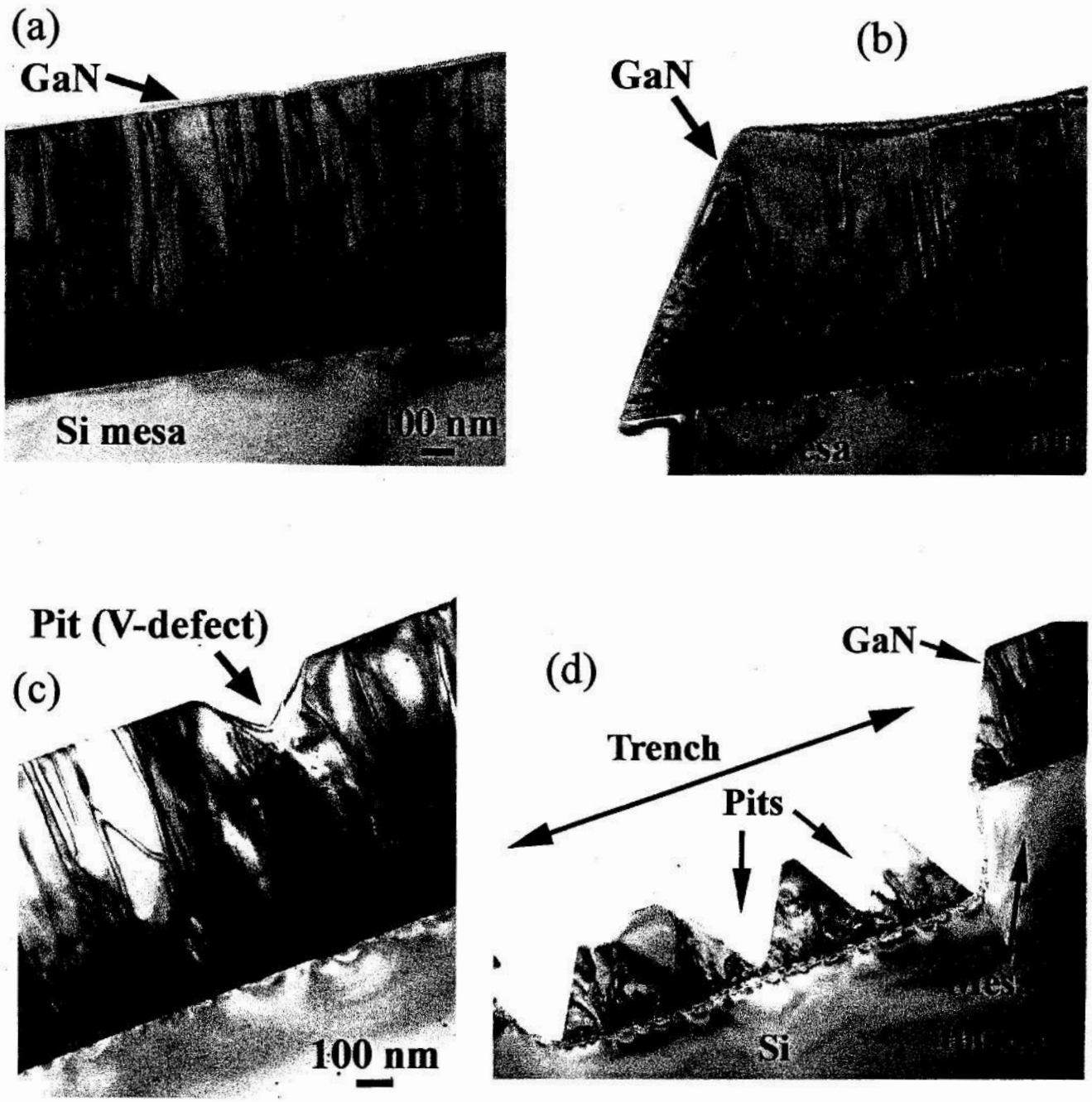


FIG. 4. Cross-sectional two-beam TEM images, taken at g = 0002, revealing the TD's in LCE GaN/Si. (a) from the center of a 20 μm pattern. TDD ~10^{10} cm^{-2}; (b) close to the edge of the pattern. TDD ~10^{9} cm^{-2}; (c) close to a V-defect in the GaN pattern with nearly singular $\{1\,\bar{1}\,01\}$ planes; (d) the trench material, having deep pits between crystallites of pyramidal shapes.

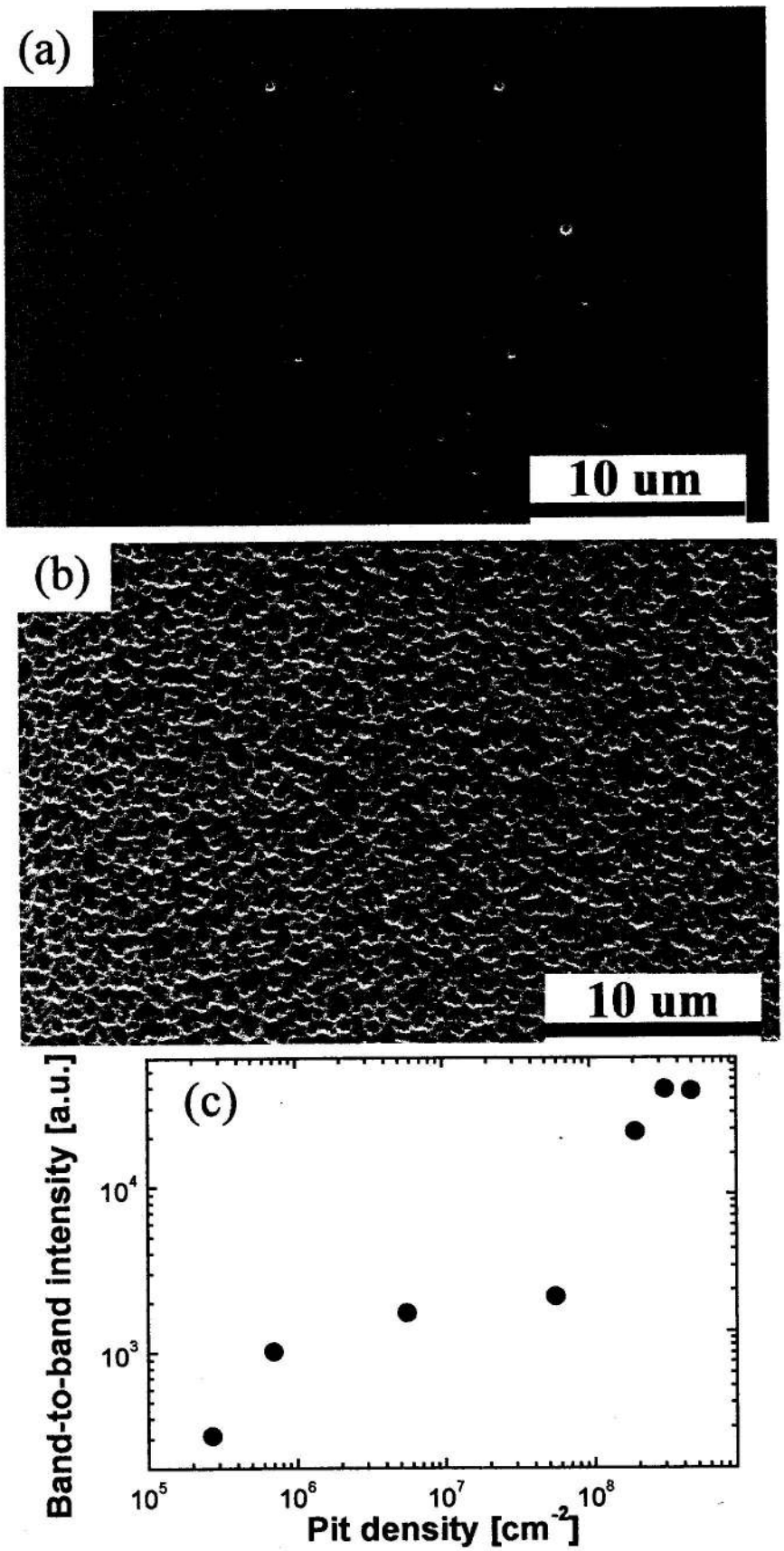


FIG. 5. Plan-view SEM micrographs of (a) the high, and (b) the low temperature areas, of the GaN/sapphire sample grown under temperature gradient. (c) The calculated pit density for different regions along the temperature gradient line, and the corresponding BB peak intensity. The PL intensity gets >200 times higher towards the high pit-density region.

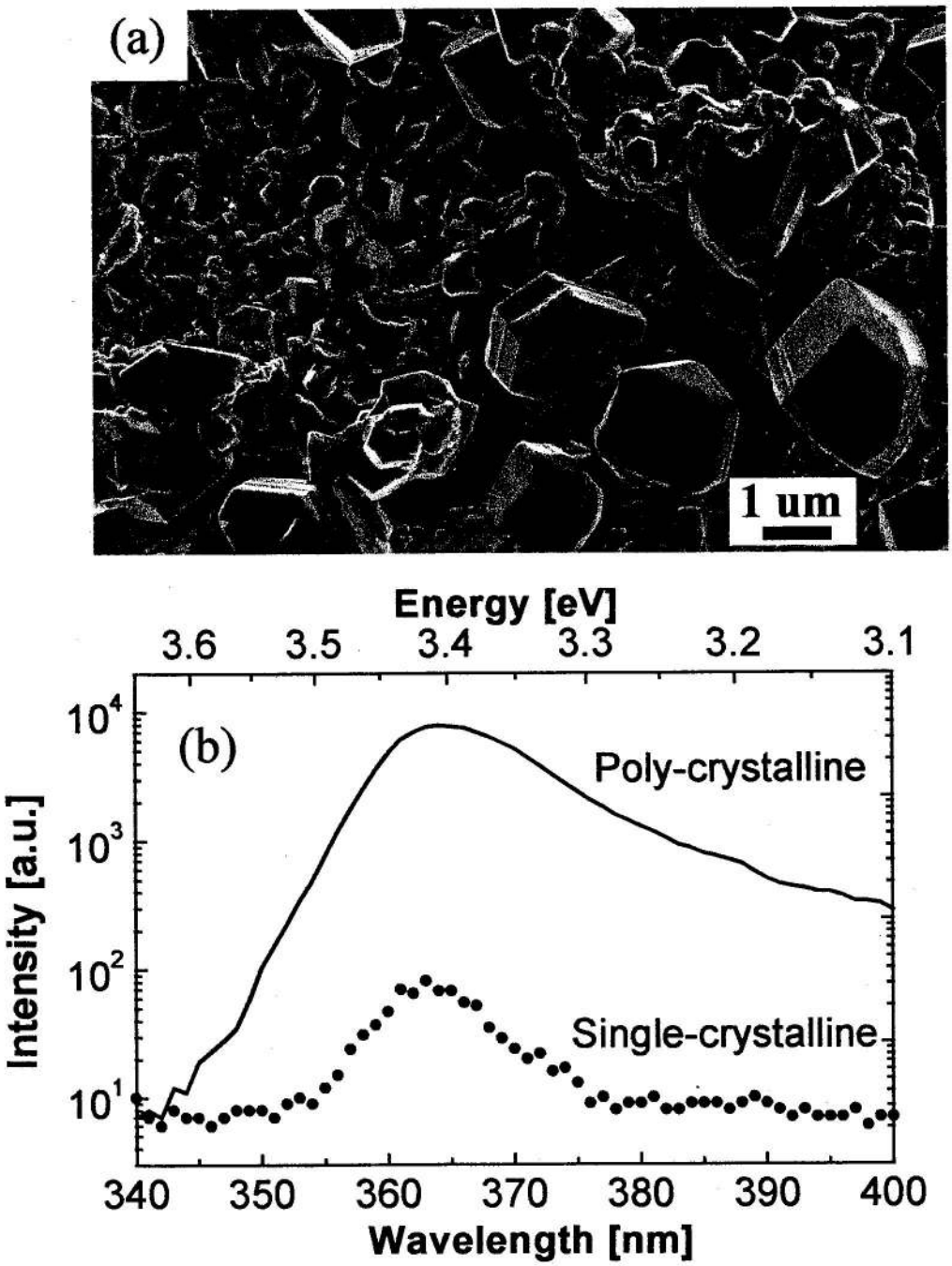


FIG. 6. (a) A plan-view SEM micrograph of a polycrystalline GaN film grown under non-optimal conditions on a Si(100) substrate; (b) a comparison between the PL intensity of this sample and that of a standard, flat, nearly single-crystalline GaN/Si sample. The BB peak intensity is ~100 times stronger for the polycrystalline film.

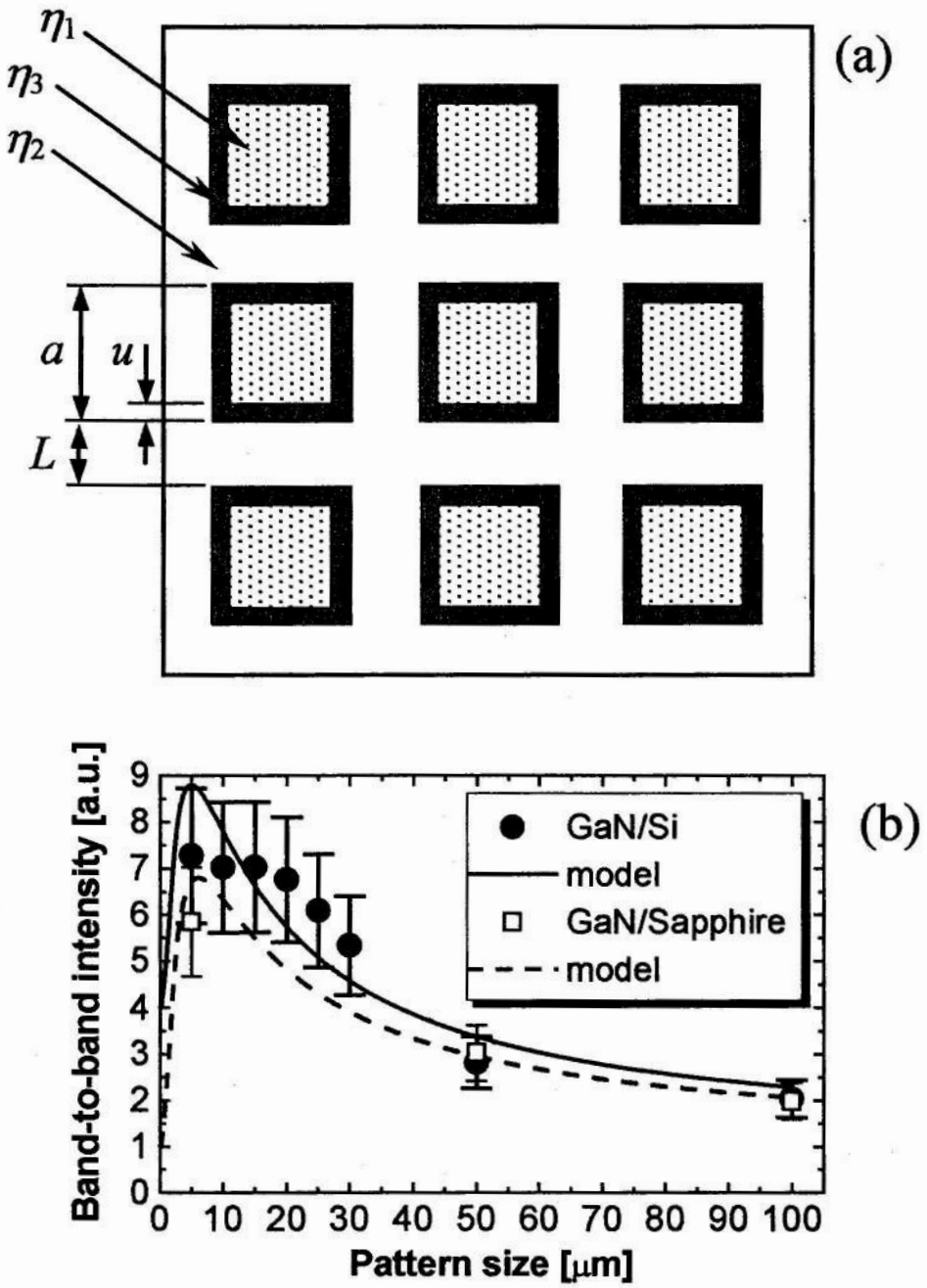


FIG. 7. (a) The different areas of an LCE sample, with their corresponding PL intensity factors, as proposed by the model; (b) a comparison of the qualitative prediction obtained from the model with experimental PL intensities for LCE GaN/Si and GaN/sapphire.

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This Technical Report is listed also as
EE PUB #1285, August 2001.